

Our Digital World — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper. Every sum is built to work by hand, using small numbers or powers of ten.
- A label alone is rarely a full answer: where a question says **explain** or **describe**, give the action or the reason. A defence only scores with its reason — “install a firewall” earns nothing; say what it does that stops the attack.
- The mark for each question is shown in square brackets on the right.
- **Fact-check checklist:** 1 Source — is it named, and can you check who they are? 2 Date — is there one, and does it fit the event? 3 Evidence — can you trace a link, the data or the full quote? 4 Hurry — is it pushing you to react or forward it? 5 Other outlets — do reliable, independent outlets report the same thing?
- **What each defence does:** a **firewall** blocks network traffic that breaks its rules; **antivirus** scans files and programs against known malware; **anti-spyware** detects and removes programs that secretly watch you.

SECTION 1 – THE VOCABULARY

1. Creative Commons licences use four short tags. In each box, write what the tag asks of the person reusing the work. [4]

Tag	What it asks of you
BY	
NC	
ND	
SA	

2. Virus, spyware and ransomware are all malware. For each one, write **one thing it does that the other two do not**. [3]

Malware	One thing only it does
Virus	
Spyware	
Ransomware	

Not “it is harmful” — all three are. Write an action the other two do not do.

3. Name the guard that does each job. Write *firewall*, *antivirus*, *anti-spyware*, or *none of the three*. [4]

The job	Guard
A downloaded attachment contains a known virus, and you want it stopped before it opens	
A stranger far away keeps trying to open connections into the school's network	
Rahul uses the password cricket1 on every account he owns, and you want that problem fixed	
A free screensaver quietly sends a list of the sites you visit to a company	

SECTION 2 – READ THE EVIDENCE

4. Class 7C has a group chat. Read what happened one afternoon.

GROUP CHAT • CLASS 7C • TUESDAY		
4:02 pm	Kabir	Quiz team is a waste of time. I am quitting tomorrow.
4:03 pm	Sneha	Seriously?? You were so excited about it.
4:11 pm	Kabir	That was NOT me. I never wrote that.
4:12 pm	Kabir	I left my account logged in on the library computer at lunch.

- (a) Name the act. [1]

- (b) Give one piece of evidence from the chat that the 4:02 message was not Kabir's own choice. [1]

- (c) Give **two** things Kabir should do next, each with a reason. [2]

5. Read this story, then the result of a quick search.

INSTAGRAM STORY · ACCOUNT @BRIGHTFUTURE_GIFTS	
THE POST	FREE LAPTOP for every Class 7 student!! Share this with 10 friends, then enter your home address and your parent's phone number at bf-gifts.example. Only 100 left. Offer ends TONIGHT.
DETAILS	Posted 3 Oct 2026 · no company named
QUICK SEARCH	No company's own website lists any such offer. The link belongs to a site registered last week. No reliable outlet reports it.

(a) For each check, write **PASS** if the post is fine on it and **FAIL** if it is not. [2]

Check	PASS or FAIL
1 Source — named, and checkable?	
2 Date — there, and fits the event?	
3 Evidence — can you trace it?	
4 Hurry — is it pushing you to react?	
5 Other outlets — do reliable ones report it?	

(b) The post asks for a home address and a parent's phone number. Explain why handing these over is risky, linking your answer to your digital footprint. [2]

6. Aadhya's cousin posts a photo of her at the school gate. The caption reads: "Aadhya, Class 7B, ready for school at 8:10 am." The cousin tags her.

(a) Does this post add to Aadhya's **active** or **passive** footprint? Give the reason. [1]

(b) Give two things a stranger could learn from the caption, and explain how **one** of them could be misused. [2]

- 7.** Zara says: “Letters make a harder password than digits, so a password of 3 lowercase letters beats one of 5 digits.” An attacker’s program makes **1,000 guesses every second**.

(a) Calculate the number of 3-letter lowercase passwords. (You are given $26 \times 26 = 676$.) [1]

(b) Calculate the number of 5-digit passwords. [1]

(c) Use your answers to show that Zara is wrong. [1]

(d) How many seconds does it take to try every 5-digit password? [1]

- 8.** Ishaan’s phone locks for an hour after **3** wrong guesses. His PIN has 4 digits.

(a) How many different 4-digit PINs are there? [1]

(b) An attacker gets 3 guesses. Write the chance that one of them is right as a fraction of all the PINs, then as a percentage. [1]

(c) Explain why locking the phone defeats a brute force attack, even though the PIN is short. [1]

(d) Name one other defence against password guessing and say what it does. [1]

End of paper. Check that every defence you named comes with its reason, and every calculation with its working.